

ActInf GuestStream #014.1 ~ Osteopathy and Mental Health (Bohlen, Esteves, Shaw)

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Hello and welcome everyone to the Active Inference Lab.

Today it is January 10th, 2022 and we're here in ActInf guest stream number 14 with a whole host of awesome guests.

We're going to be discussing osteopathy and mental health and we're going to have the authors say hello and introduce themselves.

Then Lucas will give a presentation and during the presentation, please feel free to add your comments and questions in the live chat.

And then we'll have a bunch of time after the presentation to ask those questions to the authors, hear any authors remarks.

So really excited to hear from you all and learn from your perspective.

Let's hear the introductions from the authors and then go to Lucas.

So please Robert, and then we'll kick it off from there.

Thank you.

Yeah, Robert Shaw.

I'm based in Sweden.

I'm an osteopath and a psychotherapist.

My research interest is particularly with embodiment.

And that was what my PhD was on, embodiment within the therapeutic relationship.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Hi everyone.

I'm Francesco Ceritali from Italy.

I'm a neuroscientist and an osteopath.

Mainly interest in discovering how the brain works in relation to touch, in relation to the effect of touch, based on the different type of stimuli that the body receives.

Hi, I'm George Estavils.

I'm an osteopath.

I'm currently based in Malta and Italy.

I'm also a cognitive scientist and my main interest is on predictive processing, active inference, touch and chronic pain in general.

And I'm currently working part-time at the osteopathic research Institute and I'm working there in

education and research.

And I'm also part-time studying psychology and science at mental health, mental health at King's college in London.

And I think I can kick off the presentation now if that's okay with you, Daniel.

Perfect.

Thanks.

So, yeah, we have the opportunity.

Thanks again for the active inference lab for the invitation.

It's a pleasure to be here.

We have the opportunity to present a paper that we have recently published in Frontiers in the section of psychology, the research topic called an activism and active appearance in the therapeutic alliance.

And the paper is entitled osteopathy and mental health and embodied predictive and interoceptive framework.

And yeah, without further ado, I think let's, let's dive right into it.

A little introduction on why these things might be interesting.

We have globally a very high prevalence and burden, but also comorbidity between muscle skeletal and mental health disorders.

So to put this in perspective, about 1 billion prevalent cases of mental disorders have been reported in 2017, while about 1.3 billion prevalent cases have been reported of muscle skeletal disorders in 2017.

So together these two broad spectrum of disorders are very prevalent, but also burdensome.

So together both, both conditions contribute to, are the main contributors to work by disability.

So, for example, the most skeletal mental disorders are also highly comorbid, which means for example, that anxiety, depression are frequently co-accord with chronic and often muscle skeletal pain conditions and vice versa.

So, for example, depressive symptoms are quite prevalent in pain sufferers and pain symptoms are also quite prevalent in individuals with depression.

So these, this data kind of outlines a need to, to integrate physical and mental health care services in our opinion, but also in the opinion of many research papers in the literature.

So, especially multidisciplinary collaborations between muscle skeletal and psychological health care specialists

have been called for.

But on the other hand, there have also been some disease classifications that have been proposed to, to combine physical and mental health and proposed classifications that are neither purely somatic nor purely mental.

So we believe that there is a need to integrate these kinds of approaches.

And we like to propose osteopathy to be, to be one of these approaches.

And osteopathy for those who don't know is generally used primarily to treat muscle skeletal disorders like back pain.

But it is also, has recently been proposed to benefit psychosocial factors in particular in, in, in, in patient populations with persistent pain.

And there have also, there was also some research that has been conducted from Hillary Abbey and colleagues

showing some positive outcomes of combined treatment as a, as a combined osteopathy and combined psychological

informed strategies, showing positive results.

So there are only a few attempts to, to investigate the effectiveness of osteopathy in the field of mental health.

And there are only a few approaches being developed to, to treat, or to engage with mental health.

So, based on this background, we proposed, we try to propose a framework on how to move forward and how to research the epistemic effects of osteopathy in this field.

And, uh, we do so by drawing together, um, from the research fields of embodied cognition, predictive coding, but also interoception and osteopathy.

And we like to propose, uh, in the end, our framework, which is an embodied predictive and interoceptive framework.

And if you look on the right-hand side, table one, this is kind of all our background, uh, put together and simplify body cognition kind of highlights, uh, the role of the body in cognition and mental health.

And, and these processes are, processes are likely underpinned by predictive coding, uh, processes.

So, um, philoma or access point, uh, we discussed interoception as, as an access point and therapeutic access point

to, to, uh, affect these processes.

And we argue that osteopathy might, might be an approach to, to realize this aim.

So let's have a look at the, uh, background.

We start with the first research field, which is, uh, embodied cognition.

And, um, generally embodied embodiment is an interdisciplinary field of research, which spans disciplines, uh, including philosophy,

but also psychology, psychiatry, and neuroscience.

And in general, these theories of embodiment argue that the cognition and emotion are both, uh, dependent on embedded, embodied simulations.

So they argue that body and mind are inseparable in producing cognition and that, uh, the non-neural body, uh, might both constrain and enable cognition.

But notably there are stronger and weaker versions of embodiment.

Um, some argue that the body, uh, might contribute to, uh, cognitive processes, while other versions argue that it might constitute cognitive processes.

Um, so to, uh, get a little more, a bit more detail involved, uh, theories of embodiment argue that cognition and emotion are based on the

uh, reinstatements of perception.

So this is relating to both external, exteroceptive and interoceptive, meaning interoceptive sensory states, but also to action, which relates to proprioceptive motor states.

Um, and these perceptions produce embodied or sensory motor simulations, which, uh, of previous experiences in one's self.

So this is, uh, um, kind of explanation of embodiment.

Um, and, um, this perspective is quite interesting because it might revise our brain-centered view of cognition by acknowledging that, uh, these processes are not simply limited to neural events in the brain, but also cut across the brain, body, and world's divisions and should be also studied accordingly.

And there are different kinds of commitment to the embodied, uh, theories.

And, and, and activism is an interesting, uh, aspect or linked concept that I want to, to go into detail a little bit further.

And, and activism proposed that cognition, uh, emerges from the dynamic interaction of, um, brain, body, and also environment.

So, um, most of you might have heard about the 4E cognition, um, which is that cognition is, uh, all together embodied, embedded,

and acted, extended, but also ecological.

And, um, we are, or it is argued that cognition is an embodied, embedded, enacted, extended, and ecological, um, process, uh, of sense-making, uh, through a body in an environment.

And so this is an evaluative, evaluative interaction of an organism and its environment.

So in, in detail, these four or five E's mean that, um, body cognition is enabled through, and also constrained by the non-neural body.

Um, embedded means that cognition depends on the environmental context, um, as the organism is situated within the environment.

Um, and, and acted means that cognition is for action and depends on the interaction of the embodied organism with the environment.

And extended means that cognition extends beyond the brain and body into the environment.

The most notable example might be your smartphone.

And lastly, ecological means that cognition depends on the environmental affordances for, for action.

And altogether this might constitute, uh, uh, an activism with a five E perspective.

So combining these perspectives of embodied and, uh, an active, uh, an activism, um, both mental and physical health conditions,

for example, pain and depression, um, are associated with altered interaction, uh, with an altered interaction of the organism with,

with the environment through its embodiment.

So it's, uh, the three part interaction of brain, body and environment.

So to sum this up, uh, embodiment and, and activism emphasize the interaction of brain, body and environment in understanding the mind and cognition.

And, um, to understand the, the processes of the brain in a little bit more detail, we, we have to turn to another field of research,

which is the next part of the background, which is predictive coding.

And I believe that predictive coding is, is a topic.

Most of the listeners are more familiar with than us, but I tried to summarize them anyway.

So predictive processing, um, is a theoretical framework with a growing influence in, in the field of cognitive science.

And, uh, it is, um, closely related, but also distant, uh, distinct from the free energy principle, Bayesian brain hypothesis and active inference.

So I will summarize all of them closely, uh, shortly.

So the free energy principle proposes that living systems resist a tendency to disorder, or in other words, um, remain in thermodynamic non-equilibrium steady states.

And they do so by restricting themselves to a limited number of states, uh, through the minimization of free energy.

And in, in this instance, free energy is defined as the difference between the systems predicted and actual states.

And this can be minimized using both perception and action and perception.

It is minimized by updating the prediction, uh, that is made from the generative model, uh, based on the sensation.

And action in turn, um, is changing the sensation through action to match with the prediction.

So it's like a self-fulfilling prophecy.

And, uh, the Bayesian brain hypothesis, um, proposes that brains making predictions about the causes of extra receptive,

inter receptive, but also proprioceptive sensations, um, using both the current sensory input, but also prior beliefs based on the generative model.

And, uh, in general, uh, the confidence or precision one has, uh, in the belief or the prior, um, and the sensory input or the likelihood, um, will determine how much the perception will, in the end, shift towards the expectation.

So, um, uh, in general, a gap between prior and likelihood is called a prediction error or free energy.

And, uh, while a system is minimizing prediction error or free energy or surprise or entropy, that is also maximizing the, uh, uh, evidence for its own model of the world.

And this is, uh, the Bayesian brain hypothesis is, uh, relating to behavior.

So predictive coding is putting this into a neural perspective and proposes that there are descending top-down predictions

that are, uh, conveyed from higher cortical levels down to lower cortical levels, and where they are compared to, um, us and bottom-up sensory information.

So, uh, from this perspective, information only goes up the cortical hierarchy, um,

if a mismatch between the predicted and actual information occurs, which is termed to prediction error.

And otherwise, uh, top-down predictions constantly explain away, um, bottom-up sensory information.

So, um, then there's active interference, which proposes that the brain actively modifies its input to confirm prior expectations.

And this is an active and also selective sampling of sensory information in the world through action.

So, um, prediction errors can be minimized.

We have talked about it a little bit through perceptual inference and active inference.

Uh, perceptual inference means, meaning revising the generative model based on the prediction errors and active inference, um, acting on the world to, to generate, uh, a state that is predicted by the generative model.

And to bring all of this together, uh, within the physical and mental health condition,

uh, to understand the physical and mental health conditions, um, it might be an altered weighting of prior beliefs

and sensory information, which gives rise to physical and mental health conditions.

So for example, physical health conditions, uh, prior beliefs are likely overweighted relative to sensory information,

which means I follow too much prediction, uh, precision, um, which then predicts symptoms, uh, consequently experienced.

And, um, and, um, this might also lead to heightened, um, prediction of pain, even towards harmless sensations.

So that harmless sensations might be perceived as painful.

And, uh, on the other hand, mental health conditions, um, both, uh, prior beliefs might be overly precise compare,

in comparison to sensory information or vice versa.

Um, so the first one in depression and the second one in psychosis.

Um, yes.

Uh, so one point that was, uh, quite, um, interesting is that chronic pain and depression,

which we are quite interested in, uh, are particularly linked to all experiences of interoception.

There's a little literature about this.

Um, so that's, this is our next research topic that we want to, uh, to combine with the other ones.

Um, so what is interoception?

Um, Kaiser and I, uh, have defined interoception as, uh, the process by which the nervous system senses, uh, interprets

and integrates signals originating from within the body.

So it's all information from within our own body.

And this provides, uh, some kind of embodied sensory experience, um, which is absolutely necessary for the adaptive interaction with the environment.

So, um, an interesting aspect is that, um, the active interference model was also applied to interoception, which is termed, um, interoceptive interference.

And it proposes that interoceptive experiences result from inferences about the hidden costs of this sensory information.

So to put this, um, into the neuro perspective, uh, top down interoceptive predictions, uh, compared with bottom up,

intercept, interoceptive information, most likely in the insular and the mismatch between both then results in prediction error,

which, uh, is minimized depending on the, uh, on the, uh, precision by either perceptual inference or active inference.

So the separate inference means revising the top down interoceptive predictions and active inference means then modifying the bottom up

interceptive information to convey with the prediction.

And, uh, interestingly, uh, and you said, has proposed that, uh, an emotion can also be viewed as interceptive inference because interceptive prediction errors are used to interfere emotional states.

So to, uh, connect this to physical and mental health conditions, um, deficits in interceptive processing, for example, uh, low interceptive accuracy, um, might be associated with both physical and mental health conditions

like chronic pain and comorbid depression.

um, however, there, uh, is, um, it is not clear, uh, how that every disorder in every individual might behave the same in this way.

I think there are, there are, um, differences.

So, um, so some disorders in individuals would likely benefit from reducing overly precise interoceptive predictions.

Uh, for example, robbing beliefs about real, but indeed harmless sensations, uh, while other patients might, might, uh, reduce overly precise

interoceptive information, which is, uh, for example, illusory sensations that are maintained, that maintain robbing beliefs.

So, um, there are both possibilities.

So let's talk about our approach on, uh, how to influence, uh, to, uh, interoceptive information.

And this is osteopathy.

We are all osteopaths, uh, the authors on this paper.

So a few words about osteopathy.

Osteopathy combines, uh, hands-on manual approaches with, um, hands-off patient management approaches.

So we use touch and manipulation as hands-on, uh, which is informed osteopathic models of care.

So our clinical reasoning and hands-off patient managed approaches involve, uh, patient education, psychological support, lifestyle advice,

but also, um, self-management solutions.

And these are, um, informed, of course, by osteopathic principles.

But notably both, um, might utilize top-down and bottom-up dynamics between peripheral tissues and the brain.

Um, yes.

So in general, osteopathy is applied to treat, uh, a range of clinical conditions.

Um, but, uh, an emphasis is given on chronic pain conditions and in particular musculoskeletal disorders like back pain.

And, uh, philosophically osteopathy, uh, has emphasized the unity and interaction of body, mind, and soul for, for quite some time.

Uh, and, uh, uh, it has also been proposed that, um, it might benefit psychological outcomes, especially in chronic pain patients.

But this, uh, needs some further research.

And, um, uh, Francesco and colleagues also proposed that, uh, uh, osteopathic treatment might, uh, influence interoceptive processing.

processing in a concept paper they published in I think 2016 and went on to to conduct some primary research showing that osteopathy indeed increases interoceptive accuracy but also decreases the bold response of brain correlates of interoception in patients with chronic low back pain so they do so probably by the stimulation of afferent c tactile fibers which communicate interoceptive and affective dimensions of touch so this could be relevant to both physical and mental health conditions that involve interoceptive deficits which are for example chronic pain and also depression so this is our background and I think it leads up to to the framework we like to present and the framework bridges all of these fields of research and it is an embodied interoceptive but also predictive framework and the basic basis is that both physical and mental health symptoms may result from altered precision waiting between interoceptive predictions and intercepted input so we argue that or the literature argues that either too much precision is further to the prior which is an overly precise prediction or too little precision is a further to the likelihood which is an imprecise prediction error so arguably chronic pain and depression are mainly linked to the overweighting of priors so these beliefs likely predict painful and depressive states even if the actual interoceptive information is harmless and they do so due to past experiences and they likely make these states come true through active interference so the expected symptoms might be generated through action to confirm the prediction and in order to update these prior beliefs and reduce the active interference of pain and and depressive states it is likely that surprising interoceptive input needs to be provided and to increase first of all the weight of the likelihood but also to generate prediction errors that can revise the belief issue with the predictions and thus osteopaths likely need to foster perceptual inference so actual information is used in perceptual inference to update the prediction and also the perception of these symptoms and we have a look at figure two in a moment so um this uh there's a paper by Farb uh from 2015 which which has argued that contemplative practices may alter interoceptive processing shifting regulatory habits from active to perceptual inference

and uh we believe that that osteopathy does the same so if we have a look at our main figure here um we propose that uh you see a patient in red lying on the treatment table imagine this patient has um uh chronic pain with comorbid depression and he's lying there and the osteopath provides some touch-based interventions

to the symptomatic area in this time this image the abdomen and we argue that the belief or the prior of the patient

is likely to predict physical and mental states that are associated with pain and depression because to be the likely likely

cause of uncertain information and this is the the overweighting of the prior but if we provide some treatment interventions to the patients the sensory input is not linked to these physical and mental states which is surprising but osteopathic treatment often does not involve pain but is quite pleasant if you if you touch a patient so this is a surprising mismatch of course to the patients to the patient because the expected and the actual information do not match and thus generate an intercept prediction error so these prediction errors need to be minimized and they can be minimized using active or perceptual inference and in chronic pain we argue that high precision is afforded to the belief to the prior and low precision is afforded to the sensory information and the likelihood so the first process that needs to be engaged is active interference which likely produces the symptoms that are resembling the predicted states so likely the autonomic nervous system is engaged to produce sensations that resemble the expected painful and depressive feeling so however we are in a healthcare context in osteopathic treatment so the patient might also be declined declined to to think that this is a health promoting intervention so this is not this is contradictory with the with a prediction so it's active interference might not be sufficiently to to reduce and explain the interceptive prediction errors so consequently we argue that perceptual appearances processes processes might be processes might be engaged to update them the belief and the prior based on the likelihood so the sensory information provided posture by the osteopath might update the belief

by the patient and thereby revising the generative model holding the belief and issuing the prediction so we argue that these are the processes that underpin osteopathic treatment in these kinds of patients and especially when an emotional reaction and treatment develops so yeah and in essence we argue that belief and the prediction of physical and mental states that are associated with pain and depression are updated through these interventions so the persistent and noisy interceptive predictions errors which maintain the symptoms through active interference

might be replaced with surprising and precise and might be replaced with surprising and precise interceptive prediction errors to alleviate the symptoms through perceptual infuance

so this is in a nutshell our our framework and we also of course need to discuss this framework and have a look at some implications also limitations

so in general i like to emphasize that this theoretical framework is a theoretical framework and therefore we try as experimental scrutiny

for verification of verification or falsification so future research particularly needs to explore if osteopathic treatment

indeed alleviates physical and mental symptoms in this patient population also if it actually increases interoceptive accuracy in this population and likely also decreases allostatic load and modulates autonomic nervous system activity

so there are a few measurement tools we can use for that and other limitations include

the disregard of complexity both of the health conditions but also of the therapeutic approaches so osteopathy

and mental and depression and chronic pain are all very complex conditions and we have simplified them extremely so we overlooked environmental factors and we didn't go into explaining the patient therapist

relationship

in detail as well but this framework should be understood in this context so it should be understood in the person-centered context

and future studies might also have a look at why manual treatment of peripheral tissues sometimes leads to autonomic and emotional responses

and i think this is the basic of why i think this is the basic of why i was so interested in this research and these theories

because in practice i've all frequently engaged patients which responded quite emotionally to touch

and felt quite better after so i wanted to to understand how this might happen so based on our hypothesis you can see in figure

uh three panel a that um the therapeutic input uh that is um provided during treatment may contradict the predicted

state of pain and depression and produce an interoceptive prediction error which is used to inferential emotional states

and need to be minimal minimized by active inference or by perceptual inference and what you can see in panel a

is that the initial active inference is quite quick and sharp and we have seen or i've seen and my colleagues as well

from experience that often if patients display emotional responses to in treatment they firstly experience some sympathetic activity like sweating and muscle tightness combined with emotional distress for example starting to cry

but these initial reactions frequently are replaced with a very uh parasympathetic activity so the breathing goes down and the patient

starts to relax and emotional calmness uh starts to set in and i believe this is quite well expandable with the framework we have provided here because an active inference might be the first way to engage uh in minimizing these

prediction errors and perception errors and perception of the second and more sustainable way of minimizing these prediction errors

so uh lastly you can see on panel b another idea which might come from this framework and this is the ground for multidisciplinary collaborations in the treatment of mental and physical uh disorders so we propose a kind of uh integrative interoceptive exposure therapy in in which both um physical but also mental health care specialists can come together and um work in a multidisciplinary but also person-centered care approach and the idea is if you can see on the left hand side here is that mindfulness practices

likely work top down uh helping the patient to attend to the sensation that he's or she is experiencing um but also uh affecting the sensory information like osteopathy working bottom up also um affecting the sensory information by likely producing a sensation itself and uh psychotherapy is likely to work top down to help identify and also reflect on the sensation and likely works on um works on the prior on the belief uh directly not through the sensory information so um these are a few future research ideas and now we can

conclude uh this presentation and to put it all together um this hypothesis and theory article um proposes

the embodied predictive and interest framework uh and it does so to reason and research the effect of osteopathic treatment um has putatively on comodal psychological factors in patients with persistent physical conditions and um one of the basic ideas is that persistent physical and mental conditions like chronic pain and depression um uh may be underlined by first overly precise interoceptive predictions meaning that um symptom congruent information is expected and second imprecise interoceptive prediction errors meaning that symptom incongruent information is not sensed so osteopathic treatment may provide um uncertain and surprising interceptive information to the bodily area that is associated with the physical and also comorbid mental symptoms resulting in interoceptive prediction errors that first engage active impedance processes which lead to initial autonomic activity resembling the predicted symptoms

and this is uh because it is the most likely cause of uncertain interoceptive information due to the past experiences of the patient and afterwards due to the contextual factors uh the patient likely engages perceptual inference processes that then update the maladaptive intercept predictions based on the actual interoceptive information and osteopathy might also um actually improve the ability to attend to actual uh interoceptive information so in a nutshell persistent and noisy interoceptive prediction errors that maintain the symptoms are likely replaced with surprising and precise and receptive prediction errors that arguably alleviate the symptoms and in this way uh treatment might reduce the belief about and also the prediction of physical and mental states that are associated with chronic pain and comorbid depression and we are currently also at the osteopathic research institute conducting uh research into these approaches so torsman has also joined us today um has developed an approach to psychosomatic osteopathy and we are currently having a look on how these techniques might influence stress levels in healthy participants but also patients with chronic retention type headaches or chronic musculoskeletal disorders

and we are currently having a look on how these techniques are going to be able to develop a approach to the future so that's it thank you very much for your attention i'm really looking forward to the discussion and to engaging robert george and francesco um and here are the references i think i'll scroll through them shortly and if someone is interested i think they can get the presentation is that right daniel

yes okay so that's it thanks lucas that was an awesome presentation really a lot to think and talk about so perhaps we'll first just open it up to any of the authors and colleagues here please just uh jump in wherever you'd like i think it'd be awesome to hear about just your overview or what brought you to this area of research

as well as anything that you picked up on from this last presentation

i think maybe i can say a few words because i forgot something in the beginning so um it all started with with george and

francesco um inviting inviting robert and me to join a research group that um tries to to reconceptualize osteopathic care under the active interference framework so um we have a group of osteopaths from europe um working with the active interference framework and there are also other papers uh coming uh in a special topic which is also uh has been uh edited by uh edited by george and francesco so there's

there are other ideas coming forward in the near future uh i think you know one one thing that um i guess brought us into this area you know is is a sort of a way of looking into how we can best um help our patients in particular those with um you know present with complexity um including you know chronic pain but not just running by uh you know the the broad sort of uh area of persistent physical symptoms you know those that tend to sort of um go around see different practitioners um look for a solution to their problem trying to know what's going on and you know we tend to sort of see them in practice and struggle at times if we use um you know a very reductionist approach to practice which is do a manual intervention and hope for the best you know that's that's far from um you know from what is best practice these days what is patient-centered care and also um

can you still hear i just lost george for an audio i was just gonna ask yeah me too okay

i can pick up a bit from um

there's been a kind of a problem with a with osteopathic theory over the years

um which george is uh hinting at around the reductionistic sort of bio um biomechanical model and when it comes to complexity those sorts of traditional ways of looking at people's chronic symptoms um leave something to be um well they they they they they're not really um they leave things missing we we're realizing using uh and if we draw from other disciplines that we can see that there's a potentially much better way to describe why people have these kind of chronic type symptoms and um i i came into this a bit like lucas with um being quite surprised in practice about how people react to treatment and i think we have us um it's almost a kind of a tacit uh sort of thing within the profession that we we treat people with emotional disturbance but we don't actually upfront it and so this conceptual framework is actually starting to redress the balance um which i think is really really important and trying to understand why is it people react um i think the this notion of prediction areas is um kind of an interesting one i i came into it a long time ago looking at it from a psychodynamic point of view and things called our reactions but i think what we're moving towards is trying to come up with a bit more of a unified theory and that's why i think the concepts of active inference and activism are actually starting to help and also add the participatory sense making movement also adds a lot to how we can start to understand it uh i know lucas mentioned we haven't talked about the therapeutic relationship but we are there are other papers in the pipeline that we'll look at i'm losing my voice

i hope that sort of did a bit of a summary for you george

very interesting yeah i mean uh yeah thank you george no no no i i think robert you know did a brilliant job in uh summarizing everything so you know all the challenges and the reasons why you know we we got interested in into exploring a different kind of way of of looking at things francesco or torsten would you like to add anything um otherwise anyone in the live chat is welcome to add a question and i have some questions too uh yes daniel so and one other um viewpoint that i can add at this point is that from previous research that actually we published looking at the uh and using fmri but also other uh neuroscience you know scientific tools like for example thermocamp um we ended up understand or exploring the fact that actually the brain is starting to react to specific or general touch

uh in a kind of uh similar but different way so and this actually creates a kind of a question a research question on where the ability of the specific manual therapist approach might intercede with a specific brain bottom up or top down uh in an old style way describing the how the brain works because actually now we know and looking particularly in the inactivism that uh an active inference the but the the brain actually doesn't work without the with a top down or bottom up uh bottom up stream so far but in a in a way we actually we ended up saying that probably uh the interaction with the body so using a kind of uh affordability uh affordability and sense making so creating action into the the uh into the patient and through the action of the uh the therapeutic alliance and therefore through the uh the uh professional and the operator might create a kind of a sharing uh um scenario or a sharing environment that when with the uh with the uh concept of uh predictive processing and active inference might also be explained through the mark of blankets communication and mark of blankets connection between the uh the the two uh the two elements uh so the patient and the operator that some something might happen not only through the specific activity that any type of manual therapist can do but also in within a framework of uh closeness between the um the the two uh the two elements indeed one of the of the early study that also lucas mentioned what actually we did was just touching the uh the ankles of um and healthy patient lay down on the scan and we just change our uh mental attitude on focusing towards the hand or avoiding the attention towards the hand so like mimicking the way in which a um a manual therapist was uh paying attention to what he or she is doing through manual manual touch or just being absolutely uh away from the attention of the manuals and therefore in the specific elements of the uh of the of the research was just counting a random bit beep uh through the through the headphones so and what actually we uh we saw is that in in those people that were touched uh through an attention towards the hand there there was a higher activation of the insula and the infrafrontal gyrus like in a way the the the perception of the patient

is uh is before the fact that actually is conscious of something happening from an interoceptive point of view within the body so and the body it's saying do you know what i'm feeling something that actually i'm processing

through an interoceptive framework in the in the insula and at the same time i'm changing my um endo endogenous or exogenous attention based on the way in which you touch me so and therefore this creates a more complex environment in which actually we we we would like to understand better and we think that and the inactive uh approach is one of the and with all its let's say subsections and elements that might be an interesting way of interpreting or at least trying to read what is happening in the complex clinical environment

um

thank you so i'll make an observation and then ask a question so robert i really like what you said about uh being motivated from earlier on by what you saw the outcomes of treatment were and then that was kind of alluded to in what we just heard like um you could be touching someone's ankle but it's not just the ankle that might change maybe it's their knee but maybe it's something in their head but then maybe it's something in their organ or their brain or their mind and um that idea of like entering at the anomaly is it's very clinical like i'm not that kind of a doctor but that's kind of what

i understand to be done like looking for patterns and anomalies and then it's also very at home in the realm of participatory sense making like you also addressed but also that's kind of active inference which is relative to a generative model of the world observations are either consistent or surprising in some way and so that was just kind of it's an interesting tie-in and i think it just speaks to how even in a qualitative or just a conceptual level across different domains we can like talk about active inference as well as other areas and their concordances and it's not like one has to nest over the other their different domains and um approaches so um robert any thoughts on that or i have other questions yeah i think um there needs to be some some kind of a dressing of um you got to you know you have a relationship between a practitioner and the patient and i think the participatory sense making can start to somehow dissolve a little bit of the the kind of normative language which has become problematic so so anomalies in the past have been dismissed as in oh you're not getting better because you're not uh reacting to the treatment properly whereas these types of approaches are saying oh hold on a minute there is a shared narrative and i think the narrative medicine movement is also quite important in in these ideas obviously we can't write about that as well um but it's it's it's nice i think you've picked up on that sort of multi-dimensional and crossover between different disciplines because i think that's another thing we're trying to make a big point of in this paper we need to reach out to other disciplines and say look we have this way of doing things but we know we're not we don't have the whole picture at all but if we can combine um with psychologists psychiatrists i mean all of all of us on the paper we've all gone outside of osteopathy to to look at other disciplines um and again that's the kind of another kind of it's like a meta kind of participatory sense making that goes into i think what we're trying to achieve here like someone will look at that figure and you all are saying hey we're the person above the table with the hands-on and with medical training but then there's another part that you didn't go into detail on what would be that therapy component or the niche modification component so totally agreed that the paper is like an invitation and that was really um to my question like we see in other domains and applications of active inference you reframe something that might have been perceived about like um just reward or a normative precision based approach like this is the reference range and this you're getting your objective function is to enter that reference range and it's like a red flag if it's not towards this more perhaps even customized uh reduction of uncertainty um and the body like predicting its own sensation so i just wondered what are the implications does that play a role in how the medical expert understands the scenario does it play a role in the patient's own understanding of what's happening just how beyond the research predictions and experiments that are being described how does this translate into a clinical experience so one of the things uh it's quite interesting you're bringing the you know the concept of niche construction um on the other paper that we have in the pipeline where we explore we actually argue that osteo osteopathic care is a kind of a effective inference uh you know there's an argument around

you know the construction of you know a niche uh which uh can be a diet uh in the case in most adult interactions in clinical practice but it can also go beyond that and be even a triadic kind of relationship for example when you're treating a baby where actually it's not a two-way communication between the practitioner and the baby but it's also with the with the parents with the mother and and in fact you know the first person you need to bring you need to synchronize is probably with the mother as well because you know if you if you kind of treat the baby directly you get there so i think that there's a lot of sort of alignment a lot of synchrony um to take into account into the therapeutic encounter and you know arguably the the kind of the development of a robust uh therapeutic alliance is through that when the two people sort of sharing uh sharing a common niche but also sharing a you know common kind of uh you know mental states and predicting uh in a lot of times without any words uh what the response of the other person is is likely to be so it could be uh for some people may sort of sound a little bit uh more kind of esoteric but if we put it through a lens of you know active inference in activism and so on sense making uh you're talking about things that are pre-reflective uh you're talking about things that's just happened and uh when um even down the realms of at times we hear practitioners where our own experiences practitioners on kind of body work then you feel like at times that uh you know the patient seems to be relaxing because you've got some relaxation through your own kind of body and it's a sort of it could be the esoteric side of things but also you could think that maybe that show that demonstrates some form of alignment maybe that demonstrates an element of synchrony that can be done at sort of a more physiological level so there's a lot of um you know there's a lot of mileage in in my opinion and in exploring clinical practice uh particular in particularly in situations of with an element of complexity to explore it through them you know these lenses

yeah and also yes yeah and also uh this is also in mind with the um the most recent theoretical paper just published a few a few months ago by um antonio damasio and and uh colleagues uh where actually they were looking at the uh complexity through the um the interoceptive nervous system so what actually they called for the first time ever no uh intercepted nervous system so it's a system that includes not only the fact that actually uh we need to understand what it's going in um within our body so the old concept between quotes the old concept of craig so how do you feel but it also goes into the the fact that the body actually interacts with the environment and also the uh that beautiful issue on uh neuro or neuroscience uh based on uh interoception where actually they try to predict to uh say something new in in terms of uh interoception where the uh the the main hypothesis was that the the way in which we create interoception that is a the base for a complex clinical condition is not only related to the fact that the different brain areas and therefore the interoceptive nervous system it's creating a kind of interaction between all the autonomic and non-autonomic areas but also those areas are important for um for decoding what is happening outside the body and therefore through a sun through a descending uh primary and secondary descending mechanism that actually

they're in introduce new variables within uh within the system that uh it allows the system to detect and to understand better the complexity and through this complexity created the uh appropriate

[illegible]

of psychosomatic osteopathy in our point of view.

But it's not only that we want to interact our palpation in reaction to interception, but also to body sensations, to arousals, to emotions, to belief systems.

And from my point of view, in this case,
the osteopath is less a therapist,
but more a co-regulator by the interaction between both of them,
the awareness of the patient to his own stimulus
in different levels will increase.

And secondly, from the beginning, osteopathy was interested in interrelationships.

And this means that we are already in our own profession in a way interdisciplinary.

So it means, for example, that also we could look that peripheral hypoglykmy will lead always to a central hypoglykmy, to central glucopinia even.

So it means you have no more sugar in your brain.

So even if you would touch and you do all your psychological stuff, but by your metabolic system, you create hypoglykmy in your brain.

Even then, whatever you do, it will not have a big result.

So as more fragmented areas in the functioning of the patient's health, we can include in our treatment.

And this means basically, which was in the past, not so much dominant and more active patient.

Because in research, it will show proactivity is one of the most factors to be healthy.

And basically, in an osteopathic treatment, the patient is laying passively most of the time in the past.

So here, we have not only looking in research how osteopathic treatment can help, but even change osteopathic treatment in a more postmodern way.

So where we can help the patient even more with our hands instead, what was past treatments done in osteopathic treatment before.

So actually, on that theme of kind of origins and futures of osteopathy, I'll read a comment from the live chat and anyone else can write a chat too.

Um, good morning, my dear osteopathic brothers and sisters.

I'm an American osteopath and must leave for work now, but wanted to share a history of the Still Hildreth Hospital in Kirksville, which was an outgrowth of Dr. Still's original concept, wishing to have osteopathic applications to mental health.

There are still papers, papers by Still, which came out of that institution about effects osteopathy had on mental health conditions and early research on osteopathy and mental health from Mel Friedman, who is actually my father.

So, um, any thoughts on that?

And also just keeping in mind the guest stream is kind of a two-way street, people in active inference who are perhaps hearing about osteopathy for the first time and vice versa.

Just what are some of those principles, Lucas, that you mentioned in your presentation,

informed by osteopathic principles?

Like what are some of those principles and origins and how has it changed?

Because it was very interesting what Torsten had addressed.

Yeah.

So I think an important distinction that we have to make is between American osteopaths and osteopaths, mostly around the world, because I, for myself, I'm not a medical doctor.

So in the U.S., um, uh, the old video is, is a medical doctor, uh, in osteopathic, uh, in the osteopathic field.

So they are, um, they can treat mental health conditions and psychiatric disorders, uh, and it's regulated that they can do this.

So in Germany, for example, um, I can't treat someone with a major depression because my training was not oriented to allow me to treat that.

So it is important for us to, to emphasize that we are, um, not treating mental disorders on, on our own, but that we, um, try to, um, um, address, uh, psychological factors in people with, uh, physical conditions that are visiting us quite frequently.

So I think that that is an early distinction that I have to make, uh, between our colleagues from America and, uh, then to the point of, of the history of osteopathy, uh, we came across a few papers, um, that have argued for an osteopathic psychiatry.

I think that was, uh, involved in this paper a little bit in the earlier times.

And, um, uh, there has been some work done, but, uh, the work is, as your father has outlined quite old.

So, uh, we, we try to put some new perspectives, uh, to this discussion, I think that, uh, it's not a good answer.

I think, uh, one thing, if I may, uh, would be useful to, to add is, you know, the origins of osteopathy with Andrew Taylor Steele and, and some of, you know, his early, uh, students, including, um, John Martin, Little John, were, uh, were primarily based around, you know, new concepts, um, you know, concepts of, you know, uh, sort of, uh, unity, self-regulation that were kind of, uh, relatively new in the late 19th century.

And a lot of the work was sort of one, one interesting and important concept of concept of adaptation.

So, you know, enabling people to adapt to, uh, to their day-to-day activities, to, to adapt and interact with the environment.

Unfortunately, that was a little bit lost, uh, lost over the years.

And, and osteopathy and other forms of manual therapy, and chiropractic for, uh, musculoskeletal physiotherapy became a long time based on a kind of very biomechanical model of care, you know, looking primarily at the, as structured musculoskeletal structure as the reason for, uh, quite a, quite a lot of the, of the clinical problems.

Which, you know, in fact led to, um, Hoover and American osteopathic 19, 1963 to, to write

a very important paper.

That's not many people actually read, uh, were actually acted on, which actually called for osteopathy or osteopathic medicine to be ecological medicine, because, you know, it's not about just the theology.

It's not about sort of using a pure biomechanical, biomedical model and try to find, uh, what is the cause of the problem?

What is the pathology?

So I think, you know, the interesting thing over the years, and actually that's, um, you know, something that actually, um, you know, personally attracted me a lot to this, um, to these new ideas, uh, is actually these ideas, uh, kind of are, are very much, you know, in, a sense aligned with those initial ideas of osteopathy where everything kind of started from concept of this sort of adaptation, a central concept around the free energy principle, for example, the interactions with the environment, central conscious concepts around inactivism.

So really to think about the person in that sort of unique ecological niche and how can we as practitioners, uh, kind of help that person to, you know, help with, with self regulation, et cetera, also taking into account the knowledge we have these days about what is actually the role of the nervous system pretty much about, you know, in the role of allostasis, for example, uh, not so much about some sort of the high order cognitive stuff.

So, so I think there's, there's a lot to extract from, you know, these frameworks and, and not necessarily try to rewrite the history of osteopathy or other forms of menotherapy, but actually try things in a, in a kind of a critical and, you know, serious manner to actually sort of move forward and move away from, uh, you know, from sort of, um, you know, ideas that actually no longer make that much sense from, for example, you know, practicing entirely on a biomechanical model of, of, you know, of care.

Yeah.

I like very much what George was saying.

And basically it was not only Hoover, it was already little John and McConnell who gone in this way, what Hoover explored more, and we are on the way of them basically.

So if we nowadays look also into genetics, which was in the beginning of osteopathy, not possible, look into epigenetics, look into, uh, lifestyle factors.

And with each factor of the susceptibility of a person to get sick or get health increases.

And if we include this with our patient, we are on the way of little John McConnell and Hoover was opened up for us already in the very early times of osteopathy.

Thank you, Torsten Robert.

And then anyone else?

Yeah, I just, uh, don't want to lose sight of the fact that the paper is about recognizing that even though we may not be trained to deal with psychological interventions, an awful lot of our patients have psychological, um, problems of some or another, I think something like 40%

of primary care consultations coexist with mental disturbance.

So whether we like it or not, we're dealing with it.

And this is a, this is a way we can go.

We're not training people, not saying osteopaths should become psycho psychotherapists or psychologists, but we're saying if we raise awareness of these issues, it's going to help in our patient care.

End all.

I agree very much what Robert was saying.

And even I would go a step further.

We can't avoid to interchange with emotional being with the ANS being with cognitive levels of a person.

We only can be a little bit more competent when we treat with our patient in this areas.

And this doesn't mean that we need to be a psychotherapist in treating patients.

Robert R. It's the complex systems perspective that the entity is in the niche and that the mind and body are, uh, interrelated inseparable.

And so any model that for some sort of like, um, credentialing reason chooses to ignore that is going to be operating in peril.

Robert R. So how do we recognize the system's complexity, but also the immense amount of information, you know, you could study just the liver and there would be a lot to learn or just the lungs.

So how do we pull back and actually have a healthy relationship in this very complex setting?

Robert R. So here's a sort of, um, fun question.

Robert R. Also staying with the anomaly theme.

Robert R. I was wondering if throughout your training or experiences had come across any embodied exercises

that you think are really good entry points to the predictive body and interoception.

Robert R. So like I was thinking of having one's arms in the doorframe and trying to raise them out.

Robert R. And then when you step out, you know, the arms will raise.

Robert R. And, um, I was wondering if there was any other experiences that might be that anomaly for someone to start wondering if they were interested to look into more of these approaches that you outline in the paper

Robert R. Rather than the signal processing perception framework or some other, um, non or pre act inf model.

Robert R. So I think the literature is quite dense with examples on, on how the body influences the mind in these kinds of instances.

Robert R. For example, going back to, to struck, uh, who published the study with the pen in between the teeth, uh, showing that people find comics much more funnier

if they had a pen between the teeth, because this pen activates the psychomaticus major with this primary muscle of laughing.

Robert R. So, um, there are quite a few examples on, uh, how simple manipulations of the body framework influences, uh, cognition and emotion.

Robert R. But, uh, I don't have some specific exercises, uh, to, to give to patients, um, if that answers the question.

Robert R. Well, because also, Daniel, I think it depends from the patient's condition so far.

Robert R. So if we take into account what we, what we said up to now, so it means that, uh, each pain, uh, each patient has got its, uh, his or her own, uh, clinical treatment.

Robert R. So, uh, clinical specific condition.

Robert R. So, um, uh, going into a protocol lies, for example, uh, exercise might ended up with patient having different reactions.

Robert R. So, um, therefore, the, uh, the use of, uh, these frameworks that consider the complexity of the patient, uh, within, uh, the uniqueness of, uh, of him or of her.

Robert R. So, um, uh, uh, definitely might, uh, help us to be, uh, to have fantasy or to have imagination during the, uh, during the, uh, during the, uh, during the treatment.

Robert R. So, um, uh, the, uh, during the treatment and adapt the different possibilities that might be exercise and might be, uh, treatment procedures and might be techniques and might be approaches.

[illegible]

In fact, I'm just thinking one exercise that I tend to use, and it's nothing that I developed myself, but it's used by quite a number of practitioners to teach patients about pain and about the sensations they feel is, for example, they use the analogy to the sort of a fist. So if you ask a patient, for example, in cases of patients with chronic pain, fear of movement, you tend to find that, for example, movement tends to be quite slow and sort of restricted, not so much because there's a serious limitation in the movement, but because there's a fear of movement. They tend to use, for example, ask the patient a look, and if you make a fist and you try to move your wrist, it's going to feel uncomfortable. And that's, in a sense, how your neck is trying to move against your resisting, if you try to move, relax,

you know, it's the sensations are completely different.
So you can actually create,
if you think about these exercises that are used by,
you know, quite a lot of muscle squirtual practitioners
to actually start introducing the sort of some,
some of the sensations that we want
that might actually violate their generative model
and change things a little bit.
So that's a good way of, you know,
trying to teach the patients, you know,
this is how you should feel.
This is how you should move
and not so much like trying to move against resistance,
which happens if you make a fist
and try to, you know, move your wrist.
In general, I like very much also Francesco's commentary
because in osteopathy,
we try really to identify individualized approaches.
So there was a nice research from Amy Cuddy,
where she changed body postures
and measured the cortisol and testosterone outcomes.
But in new research, it was not confirmed
because it was too,
it was not adapted to a certain person.
Sometimes it's enough that just a slight extension
in two vertebrae can give a person
a different kind of breathing experience.
In another person,
it could be that just the face could be
as little bit released in some muscles,
like for example, Lucas was mentioned.
So I think the approaches to a patient
in an osteopathic treatment is very individualized.
And it's not a general techniques for everybody,
which could help so good as an individualized approaches
in relation to different experiences.
Thanks for all the awesome and diverse answers.
So I'll ask the question carefully, I guess,

but standardized interventions,
whether they're measured in milligrams
or educational curriculum,
lend themselves towards being scientifically tested more easily,
like a two group experiment,
people who do or don't have the diagnosis
or any other number of situations.

And it was really also just highlighted
in what Torsten added with this individuated experience.
And the dyad is unique.

There's never going to be another one like that
in the history of osteopathy
with those people in that niche interacting that way.

So just wondered what insights
from all of your different areas of expertise
start to help us address like the specificity
of the particulars of an encounter,
but also know what kinds of things should be available
for population level use or on average,
you know, are eggs good or bad?

So I think this is a really big problem in osteopathy
and research, osteopathic research in general,
is what kind of control groups are we doing?

So if we are treating a patient is a mere touch
without intention, a good control group or no intervention.

So there are many problems to the,
to researching the person-centered nature of osteopathy,
but also the manual nature of osteopathy.

And I think Francesco is a good person to speak about this
because he recently published a paper on control interventions
in osteopathy.

Maybe you want to add something, Francesco?

Thanks, Lukas.

So I think the, this is a very interesting,
question, Daniel, for three reasons.

So the first one is that
we are still within a drug-based research paradigm.
because actually during the, from 1980s, some word,

the, and then with the development of evidence-based medicine,
the majority of the research actually was drug-based research.
And therefore we can control milligrams, as you said,
we can control the type of administration
and all the elements that actually created the randomized control trial,
very controlled, very specific and so on.
However, in during the last 10 years,
all these paradigms actually start to like having some creepy things.
And in which sense?
Because actually it's,
it's very far away from the,
from the clinical practice.
And the clinical practice is based on variability
and the variable and outliers.
And the variability and outliers are important to understand
what are the clinical effects of a specific treatment.
So, indeed, during the last 15 years,
this concept of having super rigid randomized control trial,
in particular in the intervention group,
and therefore in the control group actually started to change,
or start to be molded in a way of creating something,
it's much more, much closer to the reality.
Therefore, they ended up with pragmatic clinical trial.
And pragmatic clinical trial,
those trial actually are more,
are closer to the reality,
in which there is not just one administration
in terms of, for example, drug treatment,
but actually they are taking into account
all the different elements that might influence
the effect of the treatment itself.
Indeed, when actually we administer,
for example, a drug,
but then we looking for long-term effects,
long-term outcomes in the intervention group,
then goes back home and one part of this intervention group
is very, very happy because they just got a new child,
they just won the lottery and all the positive elements.

Obviously, the long-term effects of the drug treatment is definitely better than the someone instead who, for example, had lost.

So what is the concept underneath that?

It's the concept of the research is changing and also the methodological elements underneath research are changing.

Now there are pragmatic trials, but they're also what they call the benchmarking control trial.

So there are actually are trials that are important for detecting what is happening during the clinical routine life within the hospital and then adding another intervention to the normal routine care.

Then you see what is the difference between the two pathways.

So what it's saying by that?

It's saying that it is true that it's important to control all the variables within a given intervention.

But at the same time, if we look at the cause effect and therefore efficacy, yes, it's important.

If we want to look at what is in, instead it's called effectiveness.

So the clinical effect of a specific treatment, then what you do, what you specifically do, a specific treatment or 10 grams, 10 grams more or 15 grams left actually doesn't make so that difference because of the, what we are talking here.

And also because probably those 15 grams are less are and probably will have more powerful effects.

If I'm talking to the patient with the right tone and therefore creating and inducing an increased effect based on the interaction, the narrative, the shared narratives that we are talking.

So in a way it's true, but at the same time, the recent evidence are shifting the methodological research concept towards complex intervention.

Thank you for the awesome response.

And indeed it is mirroring trends in other scientific areas to study targeted perturbations and measuring dynamical systems after perturbation or testing different dynamical models rather than just summary statistic and T test.

And then, you know, rubber stamp that.

So it is absolutely really interesting.

Does anyone else have any comments or questions or things they'd like to share?

Please, Robert.

I mean, there's quite a tradition in the nursing world to look at actual research, which is real life and real life interventions.

You know, what happens at the ground level?

And I think that's really where we are.

That's what we're saying.

You know, what actually happens in the treatment room, not what we think what happens, but actually what happens.

And this is a conceptual framework that enables that.

So almost within Act-Inp, it'd be like the difference between what we think is going to happen.

Like, well, if the two comparisons of the two different amounts had this effect on the biomarker, well, then the one that had the better biomarker effect should be used in the clinic, right?

And so that's like staying within the generative model, but then always being open to having new information and being able to have discernment to know whether it's consistent or inconsistent with our expectations, but our expectations for complex systems, even people we know very well, like they can still surprise you.

And so, but always keeping that openness to the, even the tactile, which is something that a lot of other medical modalities don't do.

So maybe the tactile 5% of the time it diverts a cohort of patients towards another diagnosis.

Like the tissue is warm, but that at the whole population level wouldn't even be statistically significant, but it might be very important for a small subset.

So it's very interesting about like kind of pointing without dwelling on some of the limitations of the current medical system, which I think most people can agree is not realizing our preferences.

And then here's actually a science informed way where we can start to look at complex interventions in human systems.

precision medicine. I mean, some people may want to live in a laboratory if that means that they could make more measurements about their body. And then there's sort of the field ecology, natural history angle. So that's very interesting of how there is like a medical, along with a mental health and ecological synthesis happening in a sense through osteopathy that now a lot of different perspectives are helping to combine typified and Lucas's very dense and informative slides.

Well, ecological is one of the E's we mentioned.

Yes. There are many E's.

And like, maybe the term inactive might not come into...

Also, it's another E.

7E.

Okay. 7E. More E's.

But it also shows how different ways of describing cross fields can prevent their synthesis. Like, again, osteopathy might not have used the term inactivism, but I mean, looking back on it, it's almost...

It's not even... To say it's definitional, it's like an understatement.

So then there's a lot of latent possibility. And then how does that get applied and how does it change the way that the art and the science are carried out, are enacted themselves?

If anyone else would like to ask a question or like, what would you say to those who are learning active inference and just hearing about what you've described for the first time? Or those maybe from the medical or osteopathic community who are learning about active inference for the first time in sort of closing?

I would say that active inference provides a way in which we can actually simplify things, you know, rather than making them more complex. I think there's a lot of, you know, people would feel scared. There's another model and there's another kind of complex thing to look into it. In fact, if you look at sort of the basic foundations of active inference, is actually things make perfect sense. And in fact, you can simplify your interaction with your patient in and become a much more effective way. You know, there's a kind of a, you know, when you get to that point where you know that things are working well, since you know, your internal states are my active states and vice versa, when I can predict things. So I think it can help us with that, can also help us to understand, you know, complexity that patients bring to the table. And you know, the paper is about mental health and osteopathy. You know, a lot of people argue that osteopathy is primarily based on, focused on musculoskeletal care. But as we've been talking about, you know, most musculoskeletal care and the most musculoskeletal conditions have an element of mental health. In a sense, the patient that can't get out, can't do their normal day to day activities, when their world stopped making sense, is likely to become depressed. So if you, by working on their, you know, musculoskeletal problem, make them feel better

because indirectly or even directly, you are treating their mental health conditions. So, so I think can help us to see the bigger picture, rather than simply the dysfunction in the body, or the brain in

fact. I definitely echo that with George. And I think what I've learned, and I've been working in this field for 30 years, is this is giving us a handle on some of the uncertainty that goes on as a practitioner. A lot of my students get really scared about how to deal with unpredictable things that occur in practice. And it can be, you know, pretty challenging, because there's a lot of complexity. And I think the big take home message I'd say is, you don't have to be the expert on your patient. And this gives us a handle. Yeah, I mean, you have a hard enough time being an expert on yourself. But to be an expert on another human being is pretty damn difficult. This gives us a model framework. It's some way it's reinventing things that may other people have said in the past, but it's, it's starting to bring it all together into a package that says, right, there's a ecological niche, or you could say shared narrative, there's a way of talking to a patient that you don't have to tell them what to do, that we can kind of work out, we can facilitate healing. And that actually is what initially got me into osteopathy.

Dr. I think this is a very interesting point, Robert. And I, it resonates with a thing that is an osteopathic principle that we try to activate the self healing powers of patients. And this implies for me that we are not the therapist who brings up healing in the patient, but the patient in itself is generating the healing. And therefore, we are not the specialists who have to understand the complexity of the patient, but to be there as a counterpart and evolve with the patient to generate a more healthy life.

I can't help but remark on Robert, what you said there about not having to be an expert. It reminded me of how like the arm doesn't have to be a theoretical coffee cup subject matter expert, it has to implement a strategy of which a broad range are going to be possible. And none of them can be pre stated, but it has to implement a strategy to get the coffee cup. And so it's just a very interesting way to tie together the transdisciplinarity with also one of the threads of inactivism and ecological psychology and active inference, which is that we're doing and thinking and doing are together rather than waiting to be subject matters on experts on everything before we are able to act in a complex niche that includes us as a part, but we're not going to get the whole picture on. Well, this was a very fun discussion. It sounds like you all are on a very cool line of research, and we really appreciate it in the lab. I hope that you'll come back and discuss more as more research occurs. Thank you. Thank you, Daniel. Thank you. Thanks a lot. Okay. Have a good day. Thanks, everyone. You too. Okay. Bye bye. See you soon.